

Shantanu Shastri

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EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
Relevant Coursework: Data Structures, Algorithms, Distributed Systems, OOP, System Design	

SKILLS

Languages: Java, Python, C++, TypeScript, JavaScript, SQL, Bash/Shell
Backend & Architecture: Spring Boot, FastAPI, gRPC, Node.js, System Design, REST APIs, Microservices, ETL
Databases & Caching: PostgreSQL, MySQL, Redis, Ignite, Azure Blob, Neo4j, Milvus, Sybase, MongoDB
Infrastructure & DevOps: Kafka, Zookeeper, Docker, Kubernetes, AWS, Azure, CI/CD, SonarQube, Prometheus

EXPERIENCE

BlackRock	January 2022 – August 2026
AI Engineer Internship	June 2026 – August 2026
Multi-Service ETL: Built a fault-tolerant backend processing 500+ exceptions/week using FastAPI, Neo4j, Milvus, PostgreSQL, and Kafka for real-time, stream-based knowledge graph ingestion. Vector-Graph Architecture: Architected a hybrid system linking Milvus embeddings to Neo4j nodes via a PostgreSQL bridge, resolving data inconsistencies and enabling structured search under 100ms. API & Orchestration: Engineered a multi-agent platform (RockAI ADK) backed by 10+ rate-limited REST endpoints handling 2K+ peak RPS, advancing Straight-Through Processing toward 100%. Evaluation & Policy: Created offline-online evaluation frameworks that track agent latency and provenance, utilizing policy optimization to process 3.5M records and handle \$14T in cash efficiently.	
Software Developer 2 - Associate	January 2025 – July 2025
System Scalability: Architected a Spring Boot data framework and TTL Azure Blob microservices, automating 1.5M+ client touchpoints and reducing infrastructure storage costs by 40%. Distributed Caching: Engineered a distributed read-through Ignite cache with LRU eviction to address high-latency APIs, scaling concurrent requests and cutting response times from 450ms to 150ms.	
Software Developer 1 - Analyst	July 2022 – December 2024
Distributed Systems: Resolved singleton bottlenecks by engineering a sharded server environment with Zookeeper leader election, distributing load and boosting system throughput to 10K+ RPS. Database Optimization: Achieved a 37% reduction in PostgreSQL query latency on tables exceeding 50M+ rows by refactoring schemas and leveraging composite indexes for faster data retrieval. Full-Stack UI: Spearheaded the end-to-end development of a responsive Angular-based UI using NgRx state management, integrating Azure's LLM APIs directly into the user workflow. DevOps Engineering: Built comprehensive Azure DevOps CI/CD pipelines with integrated JUnit/Jacoco tests, reducing time-to-market by 40% while maintaining SONAR quality scores >90%.	
Software Developer Internship	January 2022 – June 2022
Application Modernization: Re-architected monolithic Pearl/Tomcat applications into a modular Java REST environment and Angular UI, drastically reducing technical debt and deployment time.	

PROJECTS

High-Performance Modular Booking Architecture
Engineered a decoupled microservices architecture using Apache Kafka, seamlessly handling bursts of 100,000+ real-time requests with minimal latency spikes and containerized Docker deployments.
Zulip Open-Source Engine Extensions
- Built a high-throughput message engine using Tornado, Redis, and RabbitMQ to process live data streams into concise summaries with zero message loss and token usage cost optimization. - Designed a cost-efficient Topic Drift Detection system to correct conversation titles in real-time, significantly optimizing interaction models to minimize continuous token usage costs.